LAB WORKSHEET

Network Lab

A full network simulator: routers, switches, an ASA firewall, PCs, laptops, a phone, printer, access point and web server. Configure it all through an emulated Cisco IOS command line.

Getting started

1. Open the lab and click a device to open its console window. Windows can be moved and **resized** from the corner.
2. Type `?` at any point to list the commands available in the current mode, and press **Tab** to auto-complete.
3. Use the terminal colour toggle in the page header to switch between green-on-black and white-on-black.
4. Everything you type is saved in the browser, so you can close the tab and pick up where you left off. Use **Erase & restart** to wipe a device.

IOS command modes

Prompt	Mode	How you get there
`R1>`	User EXEC	Fresh login
`R1#`	Privileged EXEC	`enable`
`R1(config)#`	Global config	`configure terminal`
`R1(config-if)#`	Interface config	`interface g0/0/0`

Task 1 — Basic device setup

```
enable
configure terminal
 hostname R1
 enable secret class
 no ip domain-lookup
 banner motd #Authorised access only#
end
copy running-config startup-config
```

Task 2 — Address a router interface

```
configure terminal
 interface g0/0/0
  ip address 192.168.1.1 255.255.255.0
  no shutdown
 exit
end
show ip interface brief
```

Task 3 — VLANs on a switch

```
configure terminal
 vlan 10
```

```
 name STAFF
interface range f0/1 - 8
 switchport mode access
 switchport access vlan 10
interface g0/1
 switchport mode trunk
end
show vlan brief
```

Task 4 — DHCP, NAT & wireless (stretch)

- Create a DHCP pool so PCs get addresses automatically, then check with `ipconfig` on a PC.
- Configure the ASA / router edge for NAT so the inside network can reach the web server.
- Associate a laptop with the access point and confirm connectivity with `ping`.

Verify

- `show running-config` — confirm your settings.
- `ping` between devices to prove end-to-end connectivity.
- `show ip interface brief` / `show vlan brief` — check interface and VLAN state.

Tip: the whiteboard button (bottom-right of the page) gives you a scratch space to sketch the topology and jot addresses as you plan.